

Unit 3, Lesson 8: Comparing and Performing Operations With Absolute Value



Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.



Learning Targets

- ☐ I can compare the absolute value of two numbers.
- ☐ I can perform operations that involve the absolute values of two numbers.



3.8.1 Warm-Up: Rational Numbers in the Real World

For each of the following problems, ask a classmate for his/her explanation. Once you have received their explanation, write a summary in the space provided in the box. Your classmate should read your summary and initial that it is correct.

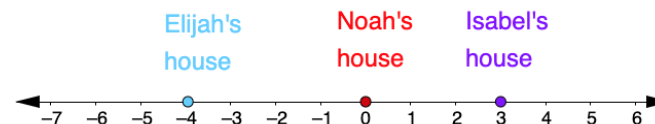
Repeat this with a different classmate for each of the remaining problems.

- | | |
|--|---|
| <p>1. Describe the relationship between two numbers that have the same absolute value.</p> | <p>2. In an American football game, the offensive team makes gains or losses on each play. Gains are represented by positive values and losses by negative values. What does 0 represent in this context?</p> |
|--|---|

Initials: _____

Initials: _____

3. Elijah, Noah, and Isabella live on the same street. The locations of their houses are modeled on the number line.



Describe where they live relative to each other in two sentences. Include the words distance and absolute value.

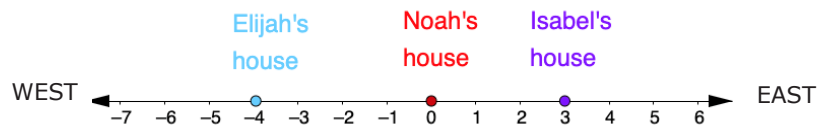
Initials: _____



3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

- Recall this scenario from the Warm-Up with some additional information.

Elijah, Noah, and Isabella live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



- Noah's house is located at 0 on the number line model. Represent the distance from Noah's house to both Isabella's house and Elijah's house using absolute value.

	Elijah's House	Isabella's House
Distance from Noah's House	$ \underline{\quad} $	$ \underline{\quad} $

- How do the absolute values in the table compare to each other?

$|\underline{\quad}| > |\underline{\quad}|$

- Isabella lives $\underline{\quad}$ kilometers to the

☐ east
☐ west

 of Noah. Elijah lives $\underline{\quad}$ kilometers to the

☐ east
☐ west

 of Noah. Isabella lives

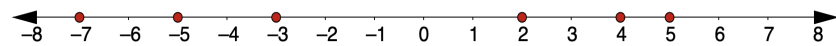
☐ closer to
☐ farther from

 Noah than Elijah does.



3.8.3 Guided Instruction: Comparing Absolute Values

A number line is shown with the integers $-7, -5, -3, 2, 4,$ and 5 plotted.



- Discuss with your partner which statements are true. Be prepared to explain your thinking.

Mathematical Statement	Explanation
$-7 < -3$	
$ -7 $ is 7	
$ -3 $ is 3	
$ -7 < -3 $	

- Does the comparison of -7 and -3 match the comparison of $|-7|$ and $|-3|$?

3. With your partner, discuss why comparing two negative values and comparing the absolute value of the same two negative numbers may be different. Summarize your discussion below.

4. For each row, use the symbols $>$, $<$, or $=$ to compare each pair of values.

	Inequality	Absolute Value Inequality
Pair 1	$4 \underline{\hspace{1cm}} -5$	$ 4 \underline{\hspace{1cm}} -5 $
Pair 2	$-3 \underline{\hspace{1cm}} 2$	$ -3 \underline{\hspace{1cm}} 2 $
Pair 3	$6 \underline{\hspace{1cm}} -6$	$ 6 \underline{\hspace{1cm}} -6 $

5. What do you notice about the comparisons?

6. When comparing the absolute values of two numbers, numbers that are closer to zero will always have the

_____ absolute value.



3.8.4 Collaboration: Inequality Mix and Match

Work with your partner to complete the table.

Partner A will select two numbers and one comparison symbol. At least one of the numbers must be written as an absolute value.

Partner B will then write the mathematical statement in words. The following phrases may be helpful: is equal to, the absolute value of, is greater than, is less than.

Switch roles until each partner has taken two turns. An example is shown in the first row.

Numbers					Symbol
-0.7	$ -9 $	11	14	$ -8 $	$<$
$-\frac{6}{2}$	-2.2	$ 2.5 $	8	$ 0.7 $	$=$
$ -14 $	$ 20 $	$\frac{7}{2}$	$ 15 $	$ \frac{5}{2} $	$>$

Partner	Mathematical Statement	Statement in Words
Example	$ 3 < 4$	<i>the absolute value of 3 is less than 4</i>
Partner A		
Partner B		
Partner A		
Partner B		

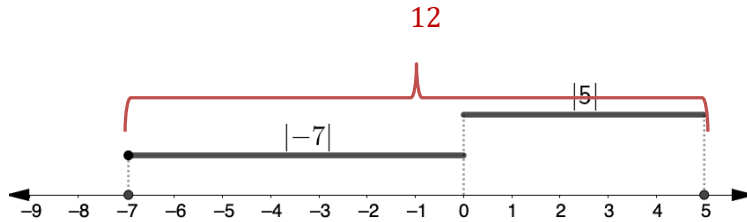




3.8.5 Guided Instruction: Operations Using Absolute Value

1. Ty'shaun and Gabby were asked to explain their strategies for determining the sum of $|-7|$ and $|5|$.

Ty'shaun modeled his thinking on the number line.



Gabby's explanation was "I know $|-7|$ is a distance of 7 from 0 and $|5|$ is a distance of 5 from 0. The sum of 7 and 5 is 12, so the sum of $|-7|$ and $|5|$ is also 12."

Explain which strategy makes more sense to you.

2. Find the value of each expression.

a. $|-18| - |-13|$

b. $12 \times |-4|$

c. $\frac{|45|}{|-3|}$

d. $|-22| + |31| + |40|$



3.8.6 Your Turn!

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

a. $|-42|$ ____ -25

d. $|-512|$ ____ 284

b. -21 ____ $|18|$

e. $|-176|$ ____ $|176|$

c. $|-582|$ ____ $|-790|$

f. $|-38|$ ____ 0

2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.

- a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin

- b. Use an inequality symbol, $<$ or $>$, to write an inequality comparing the distance each sea animal is from sea level.

$$| \quad | \quad | \quad |$$

- c. What does this inequality statement indicate about the locations of each sea animal?

3. Find the value of each expression.

a. $|-58| + 12 =$ ____

b. $|-13| \times 5 =$ ____

c. $\frac{|-36|}{|9|} =$ ____



3.8.7 Wrap-Up: Ticket Out the Door

1. The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	-132
Laguna del Carbón, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	____ ____ ____
Fort Myers, Florida	

Britton Hill, Lakewood, Florida	____ ____ ____
Laguna del Carbón, Argentina	

Laguna del Carbón, Argentina	____ ____ ____
Vpadina Kaundy, Kazakhstan	

Unit 3, Lesson 9: Operations With Absolute Value in the Real World



Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.



Learning Target

- ☐ I can perform operations that involve the absolute value of two numbers.